Syllabus Year: 2024-2025

CHRIST (Deemed to be University), Bengaluru - 560029

End Semester Examinations - November 2025

Bachelor of Computer Applications SEMESTER - I

Course Code: BCA103-1 **Max Marks: 50**

Course Name: PRINCIPLES OF SOFTWARE DEVELOPMENT - I **Duration: 02:00**

General Instructions:

- All rough work should be done in the answer script. Do not write or scribble in the question paper except your register number.
- Verify the Course code / course title and number of pages of questions in the question paper.
- Make sure your mobile phone is switched off and placed at the designated place.
- Malpractices will be viewed very seriously.
- Answers should be written on both sides of the paper in the answer book. No sheets should be detached from the answer book.
- Answers without the question numbers clearly indicated will not be valued. No page should be left blank in the middle of the answer book.

Course Outcome: Students will be able to:

CO1: Design and Develop ethical software product.

CO2: Apply terms, concepts, and tools of relational database management systems.

CO3: Construct database projects using DDL and DML queries.

Section A **2*5=10**

1)	Analyze the differences between entity sets and attributes in a database, providing examples to support your explanation.	**2**	(CO1, L4)
2)	Describe the different layers involved in a relational database management system (RDBMS).	**2**	(CO2, L2)
3)	Analyze the differences between DDL and DML queries in database management, highlighting their primary functions and usage.	**2**	(CO3, L4)
4)	Analyze the differences between entity sets and attributes in a database management system, providing examples to support your explanation.	**2**	(CO1, L4)
5)	Describe the difference between entity and relationship in the context of relational database management systems.	**2**	(CO2, L2)

Section B **5*4=20**

6)	Construct a relational database model for a library management system using DDL statements. Define at least three tables and describe their relationships through appropriate primary and foreign keys. Additionally, write two DML queries to demonstrate how to retrieve information from these tables.	**5**	(CO3, L3)
7)	Construct an example scenario where the quality assurance phase was compromised during requirement elicitation and describe how this could lead to potential problems in the final software product. Demonstrate your understanding by outlining specific steps that should have been taken to avoid these issues.	**5**	(CO1, L3)
8) a)	Analyze the challenges that may arise during requirement elicitation and suggest strategies to mitigate these issues.	**5**	(CO2, L4)

OR

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b)	Analyze the steps involved in requirement quality assessment and discuss how potential issues can arise if these steps are not properly followed during the elicitation phase.	**5**	(CO2, L4)
9)	Analyze the differences between the quality assurance phase and the requirement elicitation phase in database project development. Provide an example to illustrate your explanation.	**5**	(CO3, L4)

Section C **10*2=20**

10)	Examine the concept of entity in the context of relational database management systems (RDBMS). Analyze how different types of entities are represented within the relational model. Differentiate between various roles played by workers involved in maintaining and managing databases, and discuss their significance in ensuring data integrity and security. Finally, illustrate the creation, insertion, and deletion of a view using specific examples to demonstrate its utility.	**10**	(CO1, L4)
11) a)	Construct a relational database model for a University Library System, considering entities such as Books, Authors, Borrowers, and Loans. Describe how the relationships between these entities can be represented using primary keys and foreign keys. Additionally, demonstrate the creation of a view that provides a summary of the number of books borrowed by each borrower in a given month. Discuss any assumptions you make about the data requirements for this system.	**10**	(CO2, L3)
	OR		
b)	Construct an explanatory diagram showing the different layers of a software application and explain their significance in terms of data flow and interaction. Additionally, demonstrate how a relational database management system (RDBMS) fits into this layered architecture and discuss its importance at each layer. Your answer should cover a minimum of four layers of the software stack.	**10**	(CO2, L3)

Revised Bloom's Taxonomy (RBT) Levels

L1 - Remember
L2 - Understand
L3 - Apply
L4 - Analyse
L5 - Evaluate
L6 - Create

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